

THE STARSHIP GALILEO

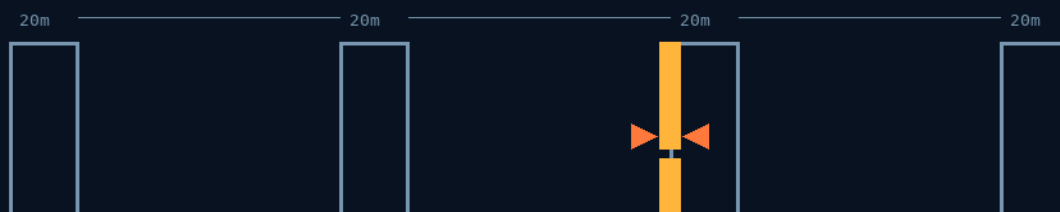
DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

PHASE 19

THE QUINN BULKHEADS. OPEN-SIDED INTERLOCKING HABITATION DECKS & HORIZONTAL BLAST GATES

Double-sided split armor — same as the outer hull
18-metre intervals — retract horizontal — symmetric deadbolts



DOUBLE-SIDED SPLIT ARMOR — SAME AS THE OUTER HULL

POD POD POD POD POD POD POD POD POD POD POD POD

BULKHEAD DOORS EVERY 20 METRES, ROUGHLY — THE CORRIDOR IS 6 METRES

GP-IM-019

Revision v1.1 — 21 August 2026

20 of 290

VETTED: PASS — RAPID COMPARTMENT ISOLATION

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPhD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 19

PHASE 19: THE QUINN BULKHEADS. OPEN-SIDED INTERLOCKING HABITATION DECKS & HORIZONTAL BLAST GATES — STATUS: CURRENT. The ship's compartment spine: primary crew decks engineered with sliding, interlocking structural gates that snap shut horizontally to seal a localized hull breach without dropping pressure in the neighbouring sub-decks. The gates are hull-grade — double-sided split armor matching the Phase 13 plating spec — retracting horizontally into bulkhead pockets at 18-metre intervals, locked by two-way symmetric interlocking deadbolts. Vetted: PASS — rapid compartment isolation; the millisecond closure is demoted to TEST/SPECIFICATION and rides §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-019, v1.1 — 21 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bulkhead law as seated (contents line; the verbatim record — rhythm, gate spec, pod-row layout) — §2 why it works (pressure boundaries, graded) — §3 the system on the ship — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Engineers the primary crew decks with sliding, interlocking structural gates capable of snapping shut horizontally within milliseconds to seal off localized hull breaches without dropping pressure in neighbouring sub-decks.' [*VET CONSTRAINT, seated §5: the millisecond figure is TEST/SPECIFICATION, not certified — the bench owes the number.*]

THE RHYTHM LAW (verbatim, the author's placement): 'bulkhead doors are every 20 metres roughly' — the rough law; the exact in-file figure is 18-metre intervals for the blast gates (below). Two nearby rhythms — doors at 18, the Phase 228 magnetosphere ring coils at 20 — live side by side; reconciling them is the author's ruling to speak (§9).

THE GATE SPEC (verbatim, from the vessel-scale block, in the file since the beginning): 'Segmented Blast Gates: Corridors feature double-sided split armor doors at 18-metre intervals that retract horizontally into bulkhead pockets. Doors feature two-way symmetric interlocking deadbolts and match Phase 13 hull plating specs.'

THE POD-ROW LAYOUT (verbatim, same block): 'Upper Deck Cross-Section: Hull pod -> 3 Accommodation pods -> 6m corridor -> 2 public utility pods -> 6m corridor -> 12 accommodation pods -> 6m corridor -> 2 public utility pods -> 6m corridor -> 3 accommodation pods -> Hull pod. Accommodation cubes arranged in blocks of 6 (3 deep x 2 wide) with dedicated side access corridors.'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PHYSICS: this is straightforward compartmentalisation — the vet's own grade. The governing line: pressure differential + intact pressure boundary = force on the gate. A breach does not have to depressurise the ship if the pressure boundary can be subdivided; the gates subdivide it.

THE MILLISECOND CLAIM, DEMOTED: a large pressure door has mass, inertia, seals, locking mechanisms and a finite travel time — 'millisecond closure: TEST/SPECIFICATION, not yet proven' (the vet, seated). Rapid automatic isolation: PASS. The number rides the bench (§9).

THE PROJECTILE CAUTION, SEATED: the gate must be designed so that pressure differential neither prevents it closing nor turns the door into an uncontrolled projectile. Driven closure with retained guidance — the door is captive in its pocket track at all times.

THE ARMOR LOGIC: a blast gate that isn't hull-grade is a planned weak point in every compartment line — so the gates match the Phase 13 hull plating spec, seated. Double-sided split armor: the gate fights from both faces.

§3 — THE SYSTEM ON THE SHIP

- THE GRID: gates at 18-metre intervals on every corridor — ~28 gate lines per 500-metre deck run (25 if the author's 20-metre rhythm ruling supersedes — his to speak, §9).
- THE DOOR: double-sided split armor, retracts horizontally into the bulkhead pocket, two-way symmetric interlocking deadbolts, Phase 13 plating spec.
- THE FAILURE-MODE CHAIN: the heavy-duty drives ride an internal slide rail PERPENDICULAR to door travel and in emergency detach and move sideways clear of the runner, so a dead drive never traps the door; if the centre lock jams, BOTH leaves slide one way and access comes via the open end; two mechanical winders (door face + adjacent room) release the drive-lock grips and move the drive with no power at all; pressure equalization (under-floor array to tanks) serves normal operation only —

emergency access never waits on it.

- THE STATIONS: Phase 163 Bulkhead Wall Stations (pod medical kit, O2 breather bottles, CO2 extinguisher, both sides of the doors) recur at every gate line — the gate line is the ship's emergency furniture line.
- THE CORRIDORS: 6 metres wide, seated (the corridor-width fill-list item is CLOSED); pod rows per the verbatim cross-section — 96m of the 100m width accounted, the 4m remainder his call (§9).
- THE BRIDGE DOORS: Phase 51 puts the same Phase 19 split armor doors on the bridge evacuation conduit, with 5-second proximity auto-closure and deadbolt engagement.

INTEGRATION NOTE (audit, flagged): the ground-test ship confirmation rides the record — the Gemini set-3 conversation PDF confirms the ground-test ship uses 'the identical Phase 19 interlocking cube pods, 6-metre corridors, and 18-metre segmented horizontally retracting blast gates.' Earth-side testability is built in.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE VETERAN PAYOUT (seated law): 'Retirement at the 20-year marker entitles the veteran to 8 fully outfitted Phase 19 pods, transported to the destination of choice to instantly construct an interlocking estate or business module.' — the pod is a pension denomination.
- THE EMERGENCY GRID : equipment is 'flush-mounted inside dedicated emergency recess fixtures on the interior cabin walls of every Phase 19/32 standardized pod' — the pod wall is a standard fitting plane.
- THE CARGO LINE : un-pressurized Phase 19 cargo pods carry mining-grade prills (Phase 128) — the same cube, unpressurized, works freight.
- THE TOILET NESTING (seated arithmetic): toilets at the 100-metre rhythm land on every 5th bulkhead line under the 20m figure — under the exact 18m gate figure the nesting no longer lands clean; flagged, not forced (§9, his ruling).

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the gate leaf: double-sided split armor to the Phase 13 plating spec (GP-IM-013 v1.4) — the leaf is a hull panel that moves, build it like one.
- STEP 2 — Build the pocket and track: horizontal retraction into the bulkhead pocket; the leaf is

captive at all times — no free projectiles under differential (§2).

- STEP 3 — Fit the drive and the deadbolts: driven closure with two-way symmetric interlocking deadbolts; power-failure behaviour defined and benched (§9).
- STEP 4 — Set the grid: 18-metre intervals per the seated spec; gate lines carry the Phase 163 wall stations both sides; corridors at 6 metres.
- STEP 5 — Commission the closure logic: auto-close on breach detection, helm authority, local manual override; the Phase 51 bridge doors take the 5-second proximity auto-closure profile.
- STEP 6 — Bench the closure number (§9): mass, drive, seal, travel time — then and only then may a milliseconds figure be spoken.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 13 — the 94mm stack (GP-IM-013 v1.4): the gates match the hull plating spec by law; the stack's chapter is the gate's materials reference.
- PHASE 228 — The Skeletal Magnetosphere: ring coils 'every 20 metres at a bulkhead' — the coil stations and the 18m gate lines are the two rhythms awaiting his ruling (§9).
- PHASE 163 — Bulkhead Wall Stations: the medical/O2/CO2 emergency furniture recurring at every gate line.
- PHASE 51 — the SMOT bridge evacuation conduit: same split armor doors, 5-second proximity auto-closure, deadbolt engagement.
- PHASE 6 & 8 — the 500m layout & 3D sliding matrix (GP-IM-006 v1.4): the pod grid the gates subdivide; the Clean Prism law the grid lives in.
- PHASE 32 — the standardized pod: the 19/32 pod wall is the standard fitting plane (§4).

§8 — DO-NOT-BUILD

- NEVER spec a gate below hull grade — a weak gate is a planned weak point in every compartment line (§2).
- NEVER certify milliseconds until the bench seats the number — TEST/SPECIFICATION stands (§5, §9).
- NEVER let differential jam or launch the door — captive track, driven closure, benched power-failure behaviour (§2, §6).
- NEVER move the rhythms unilaterally — 18 vs 20 is the author's ruling, not the builder's convenience (§9).

§9 — OWED

THE BENCH, OWED: the closure-time bench (leaf mass, drive authority, seal drag, travel time — the milliseconds number, if it survives); power-failure closure behaviour; deadbolt shear ratings; the 18m-gates vs 20m-coils rhythm ruling (his to speak); the 4m cross-section remainder (his call); toilet-line nesting under the 18m figure (flagged, not forced). All owed items ride the bench until spoken. *[PARTIALLY ANSWERED 21 AUGUST 2026 — 'power-failure closure behaviour' is now ANSWERED by the failure-mode chain (drive disengagement, centre-lock bypass, dual winders — §3/§5). The owed bench gains the SITTING 425 engineering list: transverse slide sizing; re-engagement lockout; positive cleared-runner indication; manual force at maximum differential; winder accessibility; the full sequence physically tested with pressure in both directions.]*

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-019, v1.1 — CURRENT, seated law, master v2.1 (numerical print order, post-split). Built 20 August 2026

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the §9 benches, the rhythm ruling, or his word.

END OF CHAPTER 19